We can map a few of these arrangements as  (5 rows/6 columns) and  (6 rows/5 columns) and see the opposite variations in rows and columns.

In this case, we find that the proportionality constant, k is 30.

Few more examples of Inverse Proportion:

1. Price – Consumption: If the price of consumable products increases, then naturally its consumption will decrease and vice-versa.
2. Workers – Time: If more workers are employed to complete a work, then the time taken to complete the work will be less and vice-versa.
3. Speed – Time (Fixed Distance): If we travel with less speed, then the time taken to cover a given distance will be more and vice-versa.

Now, use the concept of direct and inverse proportions and try to answer the following questions:



Try these

1. Classify the given examples as direct or inverse proportion:
   (i) Weight of pulses to their cost.
   (ii) Distance travelled by bus to the price of ticket.
   (iii) Speed of the athelete to cover a certain distance.
   (iv) Number of workers employed to complete a construction in a specified time.
   (v) Area of a circle to its radius.
2. A student can type 21 pages in 15 minutes. At the same rate, how long will it take the student to type 84 pages?
3. If 35 women can do a piece of work in 16 days, in how many days will 28 women do the same work?

Let us now see what a Compound Variation is? There will be problems which may involve a chain of two or more variations in them. This is called as **Compound Variation**. The different possibilities of two variations are:

Direct-Direct, Direct-Inverse, Inverse-Direct, Inverse-Inverse.



Note

There are situations where neither direct proportion nor indirect proportion can be applied. For example, if one can see a parrot at a distance through one eye, it does not mean that he can see two parrots at the same distance through both the eyes. Also, if it takes 5 minutes to fry a vadai, it does not mean that it will take 100 minutes to fry 20 vadais!

Let us now solve a few problems on compound variation. Here, we compare the known quantity with the unknown (x). There are a few methods in practice by which problems on compound variation are solved. They are given as follows:

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